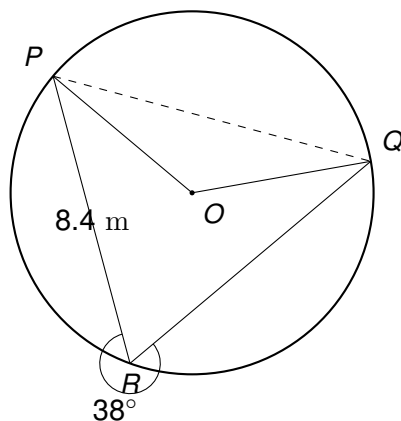


- 1 A satellite sensor rotates in a circular orbit path around a central control point O , radius 8.4 m. Two calibration markers, P and Q , are fixed on the orbit path. A tracking beacon at point R , also on the orbit, monitors the markers.
- Angle $PRQ = 38^\circ$, where PRQ is the angle subtended by the chord PQ at the circumference.



- (a) Find the size of angle POQ , the angle subtended by the chord PQ at the centre O .

$POQ = \dots\dots\dots$

(1)

- (b) The radius O to the midpoint of PQ bisects both the chord PQ and angle POQ .
Use this fact, together with your answer to part (a), to calculate the length of the chord PQ .

$\dots\dots\dots \text{ m}$

Give your answer correct to 3 significant figures.

(3)

(Total for Question 1 is 4 marks)